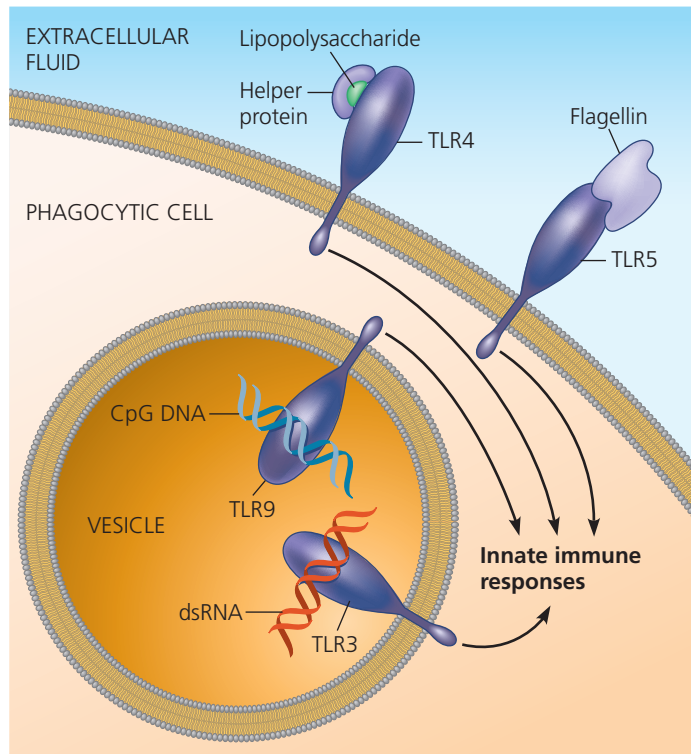
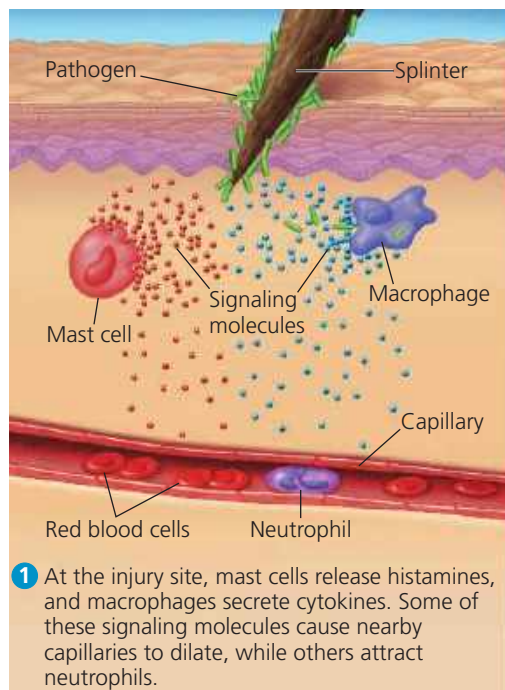


▼ **Figure 43.4 TLR signaling.** Each mammalian Toll-like receptor (TLR) recognizes a molecular pattern shared by a group of pathogens. Lipopolysaccharide, flagellin, CpG DNA (DNA containing unmethylated CG sequences), and double-stranded (ds) RNA are found in bacteria, fungi, or viruses but not in animal cells. Once bound to such a pathogen molecule, TLR proteins trigger internal innate immune defenses, including production of cytokines and antimicrobial peptides.

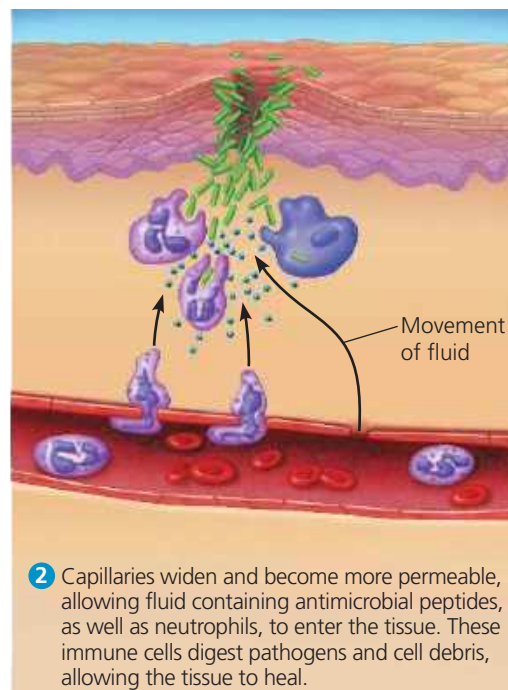


VISUAL SKILLS Observe the locations of the TLR proteins and then suggest a possible benefit of their distribution.

▼ **Figure 43.5 Major events in a local inflammatory response.**



1 At the injury site, mast cells release histamines, and macrophages secrete cytokines. Some of these signaling molecules cause nearby capillaries to dilate, while others attract neutrophils.



2 Capillaries widen and become more permeable, allowing fluid containing antimicrobial peptides, as well as neutrophils, to enter the tissue. These immune cells digest pathogens and cell debris, allowing the tissue to heal.

? From your experience with splinters, deduce whether the signals mediating an inflammatory response are short- or long-lived. Explain your answer.

➔ **Mastering Biology Animation: Overview of the Inflammatory Response**

characteristic of a set of pathogens (**Figure 43.4**). For example, TLR3, on the inner surface of vesicles formed by endocytosis, binds to double-stranded RNA, a form of nucleic acid produced by certain viruses. Similarly, TLR4, located on immune cell plasma membranes, recognizes lipopolysaccharide, a type of molecule found on the surface of many bacteria, and TLR5 recognizes flagellin, the main protein of bacterial flagella.

The two main types of phagocytic cells in the mammalian body are neutrophils and macrophages. **Neutrophils**, which circulate in the blood, are attracted by signals from infected tissues and then engulf and destroy the infecting pathogens. **Macrophages** (“big eaters”), like the one shown in Figure 43.1, are larger phagocytic cells. Some migrate throughout the body, whereas others reside permanently in organs and tissues where they are likely to encounter pathogens. For example, some macrophages are located in the spleen, where pathogens in the blood are often trapped. Macrophages and neutrophils are both key components of the inflammatory response, as discussed shortly. In addition to these cells, a number of more specialized cell types also contribute to innate immune defenses:

- **Dendritic cells** mainly populate tissues, such as skin, that contact the environment. They stimulate adaptive immunity against pathogens that they engulf.
- **Eosinophils**, often found beneath an epithelium, are important in defending against multicellular invaders, such as parasitic worms. Upon encountering such parasites, eosinophils discharge destructive enzymes.
- **Natural killer cells** circulate through the body and detect the abnormal surface proteins found on some

virus-infected and cancerous cells. Natural killer cells do not engulf stricken cells. Instead, they release chemicals that lead to cell death, inhibiting spread of the virus or cancer.

- **Mast cells** are found in connective tissue and make key contributions to the inflammatory response, described next, as well as to allergies, discussed later.

Local Inflammatory Response

When a splinter lodges under your skin, the surrounding area becomes swollen and warm. As **Figure 43.5** depicts, both changes reflect a local **inflammatory response**, a set of events triggered by signaling molecules released upon injury or infection.